

skaters: Model First, Conform Last — A Composition-Based Automatic Online Distributional Forecaster

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Abstract. The Python package **skaters** performs online univariate time-series forecasting in which every prediction is a full probability distribution rather than a point. A forecaster is built by composition: invertible transforms chain together above a single distributional leaf, and ensembles combine such chains. The leaf fits its shape by optimising a proper scoring rule, so its objective is a choice rather than a fixed property of the method. This separates two concerns the forecasting and conformal-prediction literatures tend to merge: fitting the model, judged by likelihood, and conforming the predictive tail to a downstream score such as CRPS. We call the arrangement *model first, conform last*. The library is written twice, in pure Python and in zero-dependency JavaScript that agrees to within 10^{-6} , so a model runs unchanged on a server or in a browser. We evaluate it against classical, neural, and pretrained foundation-model baselines on FRED series.

Keywords: probabilistic forecasting, time series, proper scoring rules, conformal prediction, online learning, Python, JavaScript, **skaters**.

1 Introduction

Most deployed forecasters return a point and, perhaps, an interval bolted on afterwards. Yet the quantity a decision-maker actually needs is the *predictive distribution*: the mean for an estimate, the spread for risk, the tails for position sizing, and a density for likelihood-based evaluation and combination. Two strands of recent work take distributions seriously but each gives something up.

Classical state-space and exponential-smoothing models [Hyndman et al., 2002, Hyndman and Khandakar, 2008] are distributional but rarely *online and composable*: adding a transform usually means re-deriving the filter. Conditional-heteroscedasticity models [Engle, 1982, Bollerslev, 1986] capture the heavy tails and volatility clustering of financial data, but commit to a single parametric form and are refit in batch. At the other extreme, conformal prediction [Vovk et al., 2005, 2019] is distribution-free and online, but a conformal predictive system outputs a cumulative distribution function, not a density. It is calibrated for coverage and CRPS but cannot be scored on log-likelihood at all, and an empirical conformal CDF assigns zero density — and hence $-\infty$ log-likelihood — to any value outside the observed residual range. Modern neural and foundation models [Salinas et al., 2020, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024] are powerful distributional learners, but are built for global cross-series training on hardware accelerators, not for a single univariate stream running in a browser.

skaters takes a third route. Every node — transform, ensemble, or leaf — is a function $(y, \text{state}) \rightarrow ([\text{Dist}], \text{state})$ that consumes one observation and emits a list of **Dist**

objects, one per forecast horizon, where a **Dist** is a weighted Gaussian mixture. Because the type is closed under composition, an entire model is a chain of invertible transforms with one distributional leaf at the bottom, and ensembles are simply skaters that combine other skaters. This single abstraction yields three contributions developed in turn:

1. *A pluggable objective* (Sections 3–4). The leaf fits its mixture weights by a proper scoring rule. Pointed at log-likelihood it models the conditional distribution directly; with its terminal copy pointed at CRPS while the trunk stays likelihood-weighted, the result is *model first, conform last*, which is competitive with a CRPS specialist on CRPS and stronger on likelihood.
2. *Distributional structure that survives composition* (Sections 5–7): a mean-preserving lattice projection that places atoms on the exact values a series revisits without moving the ensemble mean and vanishes on continuous data; an online coordinate search over whether a series is simple in a log, root, or linear coordinate; and a committed martingale model that learns only a volatility clock.
3. *A benchmark against eight distributional baselines* (Section 8), chosen as the strongest available opponents: the heavy-tail state of the art (**GARCH-*t***), a neural density model (**NeuralForecast**), and **AutoARIMA** paired with split- and adaptive-conformal residuals — the CRPS-optimised methods that are hardest to beat on that metric. Every method is scored on held-out log-likelihood and CRPS through a single, identical **Dist**-based protocol.

The unifying premise is the No-Free-Lunch theorem [Wolpert and Macready, 1997] taken seriously: every “method” is a prior, and averaged over all series none dominates. The right response is not to pick one but to make the priors explicit, weight them online, and bound the cost of a wrong prior by how fast the weighting abandons it.

2 The **Dist** abstraction and composable transforms

A **Dist** is a weighted mixture $\sum_i w_i \mathcal{N}(\mu_i, \sigma_i^2)$ with $\sigma_i > 0$ and $\sum_i w_i = 1$. It exposes **mean**, **std**, **pdf**, **cdf**, **logpdf**, **quantile**, and a closed-form **crps**, together with the affine maps (**shift**, **scale**, **affine**) and a moment-matching **prune** needed to propagate it through transform inverses and to bound component growth in ensembles. The Gaussian mixture is a deliberate choice: it is closed under affine maps and mixing, admits a closed-form CRPS through the elementary identity $\mathbb{E} |\mathcal{N}(m, s^2)| = 2s \phi(m/s) + m(2\Phi(m/s) - 1)$, and approximates heavy tails arbitrarily well as a scale mixture.

A *transform* is an online bijection given by a **forward** (scalar in, scalar out) and an **inverse_k** that maps k **Dists** in the transformed space back to the original. Differencing, exponential smoothing, drift, Holt’s linear trend, fractional differencing, autoregression (via online recursive least squares), **GARCH**-style scaling, seasonal differencing, power and Yeo–Johnson coordinates [Box and Cox, 1964, Yeo and Johnson, 2000], and standardisation are all transforms. *Conjugation* composes a transform with an inner skater,

$$y \xrightarrow{T} y' \rightarrow \text{inner} \rightarrow D' \xrightarrow{T^{-1}} D,$$

and because every step is online and the inverse propagates the full mixture, multi-step uncertainty accumulates correctly — for example variance grows as $\sum_h \sigma_h^2$ under differencing.

Numerical note. Mixture variance is computed in centred form, $\text{Var} = \sum_i w_i (\sigma_i^2 + (\mu_i - \bar{\mu})^2)$, rather than $\mathbb{E}[X^2] - \mathbb{E}[X]^2$. The latter catastrophically cancels when a tight component sits at a large mean, silently producing $\sigma = 0$ and an invalid predictive. Running moments use Welford’s recurrence [Welford, 1962].

3 The terminal-leaf ensemble: model first

Bayesian model averaging over a heterogeneous pool [Hoeting et al., 1999, Raftery et al., 2005] combines full predictive distributions, which preserves the combined mean and variance but *washes out higher moments*: a heavy-tailed leaf used inside the pool collapses toward a Gaussian shape at the output, because a mixture of differently located Gaussians is dominated by the spread of their means. We therefore use the candidate models only as mean forecasters, weight them online by predictive likelihood (with a learning-rate shrinkage and a per-depth complexity penalty, in the spirit of gradient boosting), combine their means, and model the distribution of the combined residual with a single terminal leaf:

$$y \rightarrow \hat{\mu} = \sum_i w_i \hat{\mu}_i, \quad r = y - \hat{\mu} \rightarrow \text{leaf} \rightarrow D, \quad \hat{D} = D \text{ shifted by } \hat{\mu}.$$

Because exactly one leaf reaches the output, its shape — heavy tails, learned online — survives undiluted. This is the *model first* half of the recipe: a likelihood-weighted trunk whose only job is to get the conditional mean right. The candidate pool spans exponential smoothing, differencing, drift, Holt’s linear trend, autoregression, grouped long-lag autoregression, fractional differencing, a seasonal block, a Yeo–Johnson coordinate grid (Section 6), and a fast/slow two-systems block.

4 Metric-agnostic leaves: conform last

The leaf is itself a fixed Gaussian *scale mixture* — components $\mathcal{N}(0, c_k \hat{\sigma})$ over a fixed log-spaced dictionary $\{c_k\}$, with $\hat{\sigma}$ an exponentially weighted scale — whose only learned quantities are the mixture weights $\{w_k\}$ on the simplex. A Gaussian scale mixture is a Student-*t* in the limit, so the leaf approximates heavy tails by construction while remaining a plain Dist. The weights are fit online by a proper scoring rule [Gneiting and Raftery, 2007]:

- **likelihood** — recency-weighted expectation–maximisation on the component responsibilities;
- **crps** — exponentiated-gradient descent on the simplex minimising the closed-form mixture CRPS [Matheson and Winkler, 1976]; the gradient is exact, since the mixture CRPS is a finite sum of $\mathbb{E}|\mathcal{N}(m, s^2)|$ terms.

Placing the CRPS leaf at the terminal of a likelihood-weighted trunk gives *model first, conform last*. The trunk’s job is to model, and the right objective for modelling is likelihood, a proper, tail-sensitive score that is, up to a constant, the Kelly log-growth criterion [Kelly, 1956]; corrupting it to chase CRPS would repeat the conformal mistake of discarding the density. The tail’s job is to be shaped, and that is the only place a downstream metric (CRPS) should get a vote. Section 9 shows this configuration improving on a pure-likelihood policy on CRPS and on likelihood alike: a CRPS-fit leaf is more outlier-robust, so it even generalises slightly better in log-likelihood.

5 The lattice projection

Administrative and grid-quoted series — policy rates, posted prices, pegged exchange rates — revisit a small set of exact values, and a continuous predictive density cannot place mass there. We add a *projection*: a recency-weighted frequency table of exact values, with near-Dirac atoms on the values revisited above a noise floor, each carrying its observed frequency as probability. The atom is mean-preserving: the continuous part is recentred so the atoms add mass without moving the ensemble mean.

Two properties make this a safe default. First, on continuous data nothing is revisited, no atom fires, and the projection vanishes identically — it is free when unused. Second, unlike a last-value spike it captures values that recur often but never consecutively (a rate that repeatedly returns to a round number). The pull of the level (its tendency to persist) is left to the trunk, where the random-walk candidate earns its weight by likelihood, so the projection remains a pure statement about mass, not mean.

6 Coordinate learning

Whether a series is simple in a linear, log/multiplicative, or root coordinate is itself learnable. We add a coarse grid of Yeo–Johnson λ candidates — the signed Box–Cox family [Box and Cox, 1964, Yeo and Johnson, 2000], defined on all of $\mathbb{R}$ — to the pool, so the ensemble learns the coordinate online rather than committing up front. This is No-Free-Lunch-safe: a wrong coordinate is simply down-weighted by likelihood. The same mechanism expresses a non-negativity prior at $\lambda = 0$ (the log coordinate), though we note that because the **Dist** is a Gaussian mixture the inverse uses the delta-method linearisation, so non-negativity is approximate rather than exact.

7 A martingale with a learned volatility clock

For near-martingale *levels* we provide a committed model, **doob**: the mean is pinned to the last value (a driftless martingale) and only the volatility clock is learned, as a Bayesian average over martingale predictives that differ in their volatility model (constant, **GARCH**, slowly varying, heavy-tailed). Because every candidate shares the same mean, plain averaging blends the clocks *without* washing out kurtosis — the committed mean is exactly what makes model averaging the right tool here, in contrast to Section 3. By the Dambis–Dubins–Schwarz theorem [Dambis, 1965, Dubins and Schwarz, 1965, Doob, 1953] a continuous martingale is a time-changed Brownian motion, so the model is literally “Brownian motion on a stochastic clock”. It beats the general forecaster on near-martingale levels by committing the mean and spending its capacity on the clock; on genuinely mean-reverting series the martingale prior is wrong and it gives ground — a sharp, single-purpose instrument, fed *levels* rather than pre-differenced changes.

The clock candidates have a Bayesian reading. Each is a *score-driven* update [Creal et al., 2013, Harvey, 2013]: the **GARCH** variance recursion $h_t = h_{t-1} + (1 - \delta)(y_t^2 - h_{t-1})$ is, term for term, the inverse-Fisher-scaled conditional score, and Hansen and Tong [2026] show via Tweedie’s formula that under a conjugate prior and local precision discounting it is the *exact* Bayesian posterior-mean correction for the variance. The same analysis identifies the smoothing factor $\alpha = 1 - \delta$ that governs the exponential-smoothing and **GARCH**-style transforms of Section 2, and recovers the Gaussian-location update as the Kalman filter. Averaging the candidates therefore blends a family of (approximate) Bayesian volatility filters, a complementary justification for treating the clock itself as the object to learn.

8 Experiments

Universe. To avoid a hand-picked series list we draw from the Federal Reserve Economic Data [Federal Reserve Bank of St. Louis], taking cached daily series and auto-transforming each to its one-step *changes* (log-difference for strictly positive levels, else first difference) — the stationary-ish innovation stream where heavy tails and regime shifts live, and which keeps log-likelihood comparable across series of different scale. We keep the last 1000 changes per series and score the last 300 as a rolling one-step-ahead test window after a burn-in.

8.1 Against eight distributional baselines

On 500 such series we compare the general forecaster **laplace** against every distributional one-step baseline we could assemble: **statsforecast**’s [Garza et al., 2022] **AutoARIMA** and **AutoETS**; **statsmodels** **SARIMAX**(1,0,1) and simple exponential smoothing, which emit *exact* closed-form Gaussian predictives; a constant-mean **GARCH**(1,1) model with Student-*t* innovations [Bollerslev, 1986, 1987] from the **arch** package, the classical heavy-tail / volatility-clustering state of the art; a **NeuralForecast** [Olivares et al., 2022] multilayer perceptron with a Student-*t* distribution head; and **AutoARIMA** paired with conformal residual quantiles [Vovk et al., 2019] in both a split-conformal and an adaptive-conformal (ACI, Gibbs and Candès, 2021) variant.

The protocol is a *fair rolling one-step-ahead*: each baseline is fit on an expanding or rolling window with periodic refit, and every method — ours and theirs — is turned into the *same* **Dist** and scored on held-out log-likelihood *and* CRPS over the same test window. The parametric Student-*t* predictives (**GARCH**, **NeuralForecast**) are represented as finite Gaussian scale mixtures, matching the analytic *t* density to $\sim 10^{-3}$. We report per-series win-rates (the fraction of series on which **laplace** scores better), and additionally restrict to the 389 continuous series — those with under 5% exactly-repeating one-step changes — since on repeating/grid series the lattice projection (Section 5) gives a large but metric-specific advantage that would otherwise dominate the aggregate. Table 1 reports the result (the unified **study.py** **sota** run).

The likelihood race. **laplace** wins the per-series log-likelihood race against every baseline decisively (85–94% of continuous series) — including the two exact-Gaussian references (**SARIMAX**, **statsmodels** ETS) for which there is no band-reconstruction approximation to blame — with a single exception, a near dead-heat with **GARCH-*t***.

GARCH-*t* is the tightest case. The **GARCH**(1,1)-*t* model is purpose-built for the heavy tails and volatility clustering of financial change-series, so it is by far the strongest opponent on log-likelihood, and on this popularity-ranked universe (heavy in rates, FX, and indices) the per-series race is a *near dead-heat*: **laplace** has the higher held-out log-likelihood on only 51% of continuous series. Its *mean* continuous log-likelihood nonetheless edges ahead, 3.10 versus 2.92 — about 0.18 nats per observation, a slightly larger gap than the win-rate suggests because **laplace** avoids the worst tail mis-specifications. That a general forecaster matches the heavy-tail specialist series-by-series, and edges it on average, without committing to a parametric volatility law, is the substantive point — stated with the honest caveat that against this one opponent it is a tie, not a rout.

The neural result, in context. **laplace** beats NF-Student*t* on 100% of series (carried from the prior run; the unified run omits NF, which requires **torch**), but the comparison

Baseline	LL (all / cont.)	CRPS (all / cont.)	Mean LL (cont.)
AutoARIMA	88 / 85 %	68 / 64 %	2.79
AutoETS	95 / 94 %	77 / 73 %	2.76
SARIMAX (statsmodels)	89 / 86 %	64 / 59 %	2.88
ETS (statsmodels)	93 / 92 %	74 / 70 %	2.87
GARCH (1, 1)- <i>t</i>	59 / 51 %	40 / 33 %	2.92
AutoARIMA + conformal	87 / 83 %	31 / 21 %	2.39
AutoARIMA + ACI	89 / 86 %	31 / 21 %	2.52
Prophet [†]	100 / 100 %	100 / 100 %	1.86
NF-Student [‡]	100 / 100 %	78 / 76 %	0.82

Table 1: Per-series win-rate of **laplace** versus each baseline on 500 popularity-ranked daily FRED change-series, all and on the 389 continuous series. LL counts series where **laplace** has the higher held-out log-likelihood; CRPS where it has the lower CRPS. The final column is each baseline’s mean continuous log-likelihood (**laplace**: 3.10). Higher is better for LL; lower for CRPS. [†]Prophet refits at every step ($n = 30$; a fresh Stan fit has no online update). [‡]NF-Student t is carried from the prior run — it requires **torch** and was not re-scored in this run.

needs reading carefully. NF here is **NeuralForecast** run in the matched online univariate one-step protocol, a small per-series multilayer perceptron with periodic refit, which is its weak regime. Neural and foundation forecasters [Salinas et al., 2020, Olivares et al., 2022] are built for global cross-series training, where they amortise structure across thousands of related series. The result says nothing about **DeepAR**-style models in that setting, and we do not claim it does; it says only that a single short univariate stream does not give a neural density head enough to beat an online ensemble. The NF densities are well formed (median log-likelihood 0.50, none floored), so the 100% is a real calibration gap, not a degenerate score.

CRPS. As ever, CRPS tells a mixed story. **laplace** beats the mean-model baselines on CRPS — the **ARIMA** family (**AutoARIMA** 64%, **SARIMAX** 59%), the smoothing models (**AutoETS** 73%, ETS 70%), and Prophet — but *loses* to **GARCH**- t (33%) and the conformal variants (21%), which are CRPS-optimised (or volatility-specialised) by construction. This is exactly the thesis of Section 4: likelihood is the metric where a faithful density wins; CRPS is conformal’s and **GARCH**’s home turf, where we are competitive but not dominant. A practitioner who truly only cares about CRPS can repoint the leaf (Section 9); a practitioner who cares about the density — for risk, sizing, or combination — should prefer the log-likelihood column.

Scope. Five hundred popularity-ranked daily change-series, one-step horizon. Multi-step horizons and raw levels are left for later; zero-shot foundation models get their own protocol in Section 8.2.

8.2 Against zero-shot foundation models

Pretrained time-series foundation models use a fundamentally different protocol: they are applied *zero-shot*, conditioning on a context window with no fitting. We evaluate four — Chronos-Bolt [Ansari et al., 2024], TimesFM 2.5 [Das et al., 2024], Moirai [Woo et al., 2024], and Lag-Llama [Rasul et al., 2023] — on 120 change-series (69 continuous). At each

Model	density	LL (all / cont.)	CRPS (all / cont.)	Mean LL (cont.)
Moirai (1.1-R-small)	native t	98 / 97 %	94 / 93 %	0.92
Lag-Llama	native t	67 / 99 %	65 / 94 %	0.39
Chronos-Bolt (small)	quantile*	76 / 100 %	67 / 88 %	−1.96
TimesFM (2.5-200M)	quantile*	75 / 100 %	58 / 72 %	−1.18

Table 2: Per-series win-rate of **laplace** versus four zero-shot foundation models on 120 change-series (69 continuous). *Chronos-Bolt/TimesFM log-likelihood is reconstructed from a quantile head and is tail-limited; read their CRPS as the fairer signal. **laplace** mean continuous logpdf: 1.51.

step the model receives a fixed 256-length window of the preceding changes and predicts the next change, with all windows for a series batched into one inference call. Each predictive is turned into the same **Dist** — the native Student- t head for Moirai/Lag-Llama, a sample/quantile reconstruction for Chronos-Bolt/TimesFM — and **laplace** is re-scored on the identical window. Table 2 reports per-series win-rates.

On the continuous series **laplace** beats all four foundation models zero-shot on both metrics (97–100% LL, 72–94% CRPS). The native-density models (Moirai, Lag-Llama) are the meaningful likelihood opponents and the closest, but still lose per-series. On repeat-heavy/grid series the foundation models are competitive or better — Lag-Llama even beats **laplace** on mean log-likelihood over the *full* universe (6.54 vs 4.52) by placing tight mass on revisited values, the same effect as the lattice projection — which is why the continuous split matters. The scope is narrow: zero-shot, no refit, fixed context, and forecasting the change stream rather than the levels these models were chiefly trained on.¹

9 Ablations: repointing the objective

A companion sweep on 10,822 FRED series (every daily-tagged series by popularity) isolates the effect of the terminal-leaf objective against the **crepes** conformal predictive system (scored on its own CDF via the pinball decomposition of CRPS). We report the single fixed **laplace** default — not a per-series best-of, which would be post-hoc selection — and give the opponent its best calibration window per series. Because this conformal opponent uses a naive (zero-change) mean, we read it as an ablation of our own objective knob rather than a state-of-the-art claim. Per-policy, de-correlating to one vote per correlated family (198 families):

Three further observations motivate the small public API. Priors wash out: across 25 series with ≥ 2000 changes the conventional prior-flavoured policies (trend, long-memory, drift, minimax, fast/slow) sit within 0.0006 nats of the uniform-prior general forecaster on the second half of each series, and the first-half winner beats the general forecaster on the second half only 44% of the time, worse than chance. Conform-last is near-free: on idealised light-tailed processes the CRPS leaf costs 0.005–0.007 nats versus the likelihood leaf, while on heavy-tailed and stochastic-volatility processes it helps (up to +0.04); real data lives in the second regime. And the lattice projection is free when unused. Together these justify exposing exactly one general forecaster (uniform prior, CRPS leaf, lattice

¹Per-series *fine-tuning* does not rescue the foundation models on this task: naively fine-tuning Lag-Llama on a single series’ history (5 epochs) *catastrophically overfits*, collapsing its held-out log-likelihood from +1.5 to below −100 while taking $\sim 15\times$ longer. This is the expected failure mode — adapting a pretrained model to one short univariate stream is not its design regime; zero-shot (or domain-level fine-tuning across many series) is the intended use. We therefore report the zero-shot result as the headline.

Forecaster (identical structure)	CRPS, raw	CRPS, family	Mean LL
likelihood trunk + likelihood leaf	92.1 %	55.6 %	2.97
likelihood trunk + CRPS leaf	96.6 %	64.8 %	2.99
bare CRPS leaf (no trunk)	92.6 %	61.0 %	2.82

Table 3: Effect of the terminal-leaf objective (fixed `laplace` vs best-of-crepes, 10,822 series). The middle row is the *model first, conform last* recipe (the `laplace` default: CRPS leaf, lattice projection on). A general likelihood policy already edges conformal on CRPS over independent families (55.6%); repointing only the terminal leaf to CRPS lifts that to 64.8% and raises log-likelihood ($2.97 \rightarrow 2.99$), matching the dedicated CRPS specialist while modelling faithfully. Crepes’ CRPS is U-shaped in window length (optimum near 250); `laplace` beats its best-of- $\{60 \dots 1500\}$ on 96.4% of series, so the gap is structural, not a calibration-window choice.

projection and coordinate learning on by default) plus one committed martingale specialist; the remaining structural ideas do not pay their way out of sample.

10 Discussion

The unifying theme is No-Free-Lunch [Wolpert and Macready, 1997] taken seriously. Every “method” is a prior; averaged over all series none dominates. The correct response is not to choose one but to make the priors explicit, weight them online, and bound the cost of a wrong prior by how fast the weighting abandons it. Read that way, the named methods of classical forecasting are convenience bundles, and the natural object is a single adaptive ensemble with two orthogonal terminal knobs — *which proper scoring rule the leaf optimises*, and *whether a lattice projection is active* — neither of which is a prior over the trunk. The empirical headline supports the design: against the strongest distributional opponents available on CPU, including the heavy-tail and neural state of the art, the general forecaster wins the likelihood race, and where it does not lead — CRPS — the deficit is confined to methods explicitly optimised for that metric and is recoverable by repointing a single knob.

11 Heritage and reproducibility

`skaters` is a from-scratch rewrite distilling ideas from `timemachines` [Cotton, 2021], an earlier competition-tested online-forecasting package, and builds on years of running live distributional-prediction contests at Microprediction, where forecasts are scored continuously as full distributions. The library is implemented twice — pure Python and zero-dependency JavaScript — and a parity suite checks roughly 90,000 probe values to 10^{-6} on every run, so results are reproducible across both runtimes and in the browser via `Pyodide` [The Pyodide development team, 2021]. The benchmark of Section 8 is a single parallel script (`benchmarks/sota_study.py`) that scores every method through the identical `Dist` interface; given a FRED key and the optional baselines (`statsforecast`, `arch`, `statsmodels`, `neuralforecast`) it reproduces Table 1 end to end.

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